

Homework 2 (of 5)

Probability and Statistics

Please Note:

1. Clearly write your student name and ID.
2. No need to repeat writing the questions, just clearly label the question number.
3. Hard copy submission during the classes of 10 or 11 October.
4. Late submission is not accepted.
5. Your assignment paper will not be returned. Please keep a copy for yourself.

1. The following contingency table summarizes the number of students at a college who have a Mastercard or a Visa credit card.

	Have Visa	Do Not Have Visa	Total
Have Mastercard	1,000	1,500	2,500
Do not have Mastercard	3,000	4,500	7,500
Total	4,000	6,000	10,000

- a** Find the probability that a randomly selected student
- (1) Has a Mastercard.
 - (2) Has a VISA.
 - (3) Has both credit cards.
- b** Find the probability that a randomly selected student
- (1) Has a Mastercard or a VISA.
 - (2) Has neither credit card.
 - (3) Has exactly one of the two credit cards.

2. Each month a brokerage house studies various companies and rates each company's stock as being either "low risk" or "moderate to high risk." In a recent report, the brokerage house summarized its findings about 15 aerospace companies and 25 food retailers in the following table:

Company Type	Low Risk	Moderate to High Risk
Aerospace company	6	9
Food retailer	15	10

If we randomly select one of the total of 40 companies, find

- The probability that the company's stock is moderate to high risk given that the firm is an aerospace company.
- The probability that the company's stock is moderate to high risk given that the firm is a food retailer.
- Determine if the events *the firm is a food retailer* and *the firm's stock is low risk* are independent. Explain.